

$$S = \underline{1 \cdot 2 \cdot 3 \cdot 4} + \underline{2 \cdot 3 \cdot 4 \cdot 5} + \underline{3 \cdot 4 \cdot 5 \cdot 6} + \dots + (\text{upto } n \text{ terms})$$

$$t_n = \overset{n-1}{n} (n+1) (n+2) \overset{n+4}{(n+3)}$$

$$= \frac{1}{5} n (n+1) (n+2) (n+3) [(n+4) - (n-1)]$$

$$= \frac{1}{5} [n(n+1)(n+2)(n+3)(n+4) - (n-1)n(n+1)(n+2)(n+3)]$$

$$T_1 = \frac{1}{5} [\cancel{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5} - \cancel{0 \cdot 1 \cdot 2 \cdot 3 \cdot 4}]$$

$$T_2 = \frac{1}{5} [\cancel{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6} - \cancel{1 \cdot 2 \cdot 3 \cdot 4 \cdot 5}]$$

$$T_3 = \frac{1}{5} [\cancel{3 \cdot 4 \cdot 5 \cdot 6 \cdot 7} - \cancel{2 \cdot 3 \cdot 4 \cdot 5 \cdot 6}]$$

$$\vdots$$

$$T_n = \frac{1}{5} [n(n+1)(n+2)(n+3)(n+4) - (n-1)n(n+1)(n+2)(n+3)]$$

$$S = \frac{1}{5} n(n+1)(n+2)(n+3)(n+4)$$

$$1 + 2 + 3 + \dots + \underline{n} = \frac{n(n+1)}{2}$$

$$1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n(n+1) = \frac{n(n+1)(n+2)}{3}$$

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + 3 \cdot 4 \cdot 5 + \dots + n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$$

differences are in g.p

Q find sum of n terms of the series $3 + 8 + 22 + 72 + 266 + \dots$ (n terms)

$$\begin{array}{l} \underline{\underline{S_0}} \rightarrow 3, 8, 22, 72, 266, \dots \\ L_2 \rightarrow 5, 14, 50, 194, \dots \\ L_1 \rightarrow 9, 36, 144, \dots \\ L_0 \end{array}$$

level 2 g.p $\rightarrow t_k = \underline{a}r^k + \underline{b}k + \underline{c}$
 $t_k = a4^k + bK + c$

$r = \text{common ratio}$
 $r = 4$

$$t_1 = 4a + b + c = 3 \quad -12a - b = -5 \quad 36a = 9$$

$$t_2 = 16a + 2b + c = 8 \quad -48a - b = -14$$

$$t_3 = 64a + 3b + c = 22$$

$$a = \frac{1}{4}, b = 2, c = 0 \quad t_k = \frac{1}{4}4^k + 2k$$

$$S = \sum_{k=1}^n t_k$$

$$= \sum_{k=1}^n \left(\frac{1}{4}4^k + 2k \right)$$

$$= \frac{1}{4} \sum_{k=1}^n 4^k + 2 \sum_{k=1}^n k$$

$$= \frac{1}{4} [4 + 4^2 + 4^3 + \dots + 4^n] + 2 \times \frac{n(n+1)}{2}$$

$$= \frac{1}{4} \times \cancel{4} \times \frac{4^n - 1}{4 - 1} + n(n+1)$$

$$= \frac{1}{3} (4^n - 1) + n(n+1)$$

$$\text{level 3 eqn} \rightarrow t_k = ar^k + \underline{bk^2 + ck + d}$$

$$\text{level 1 eqn} \rightarrow t_k = ar^k + \underline{b}$$

Q If $\underline{\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots + (\text{upto infinity})} = \frac{\pi^2}{6}$, then find value of

$$\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots + (\text{upto infinity}) = ?$$

given

$$\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \dots = \frac{\pi^2}{6}$$

$$\Rightarrow \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) + \left(\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \frac{1}{8^2} + \dots \right) = \frac{\pi^2}{6}$$

$$\Rightarrow \left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right) + \frac{1}{2^2} \left(\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \right) = \frac{\pi^2}{6}$$

$$\Rightarrow \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots + \frac{1}{4} \times \frac{\pi^2}{6} = \frac{\pi^2}{6}$$

$$\begin{aligned} \Rightarrow \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots &= \frac{\pi^2}{6} - \frac{\pi^2}{24} \\ &= \frac{4\pi^2 - \pi^2}{24} \\ &= \frac{\pi^2}{8} \end{aligned}$$

Quadratic eq

$$ax^2 + bx + c = 0, a \neq 0$$

roots $\rightarrow$ splitting

$\rightarrow$ quadratic formula

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$\rightarrow$ value putting

Q Find real roots of the eq

a) $x^4 + 3x^2 - 4 = 0$

b) $2^{2x} + 2^{x+1} - 3 = 0$

c) $(x+1)(x+2)(x+3)(x+4) = 24$

d) $(a+b-2c)x^2 + (b+c-2a)x + (c+a-2b) = 0$

e) $x^4 + 2x^3 - 6x^2 + 2x + 1 = 0$

$$a) x^4 + 3x^2 - 4 = 0$$

$$x^2 = y$$

$$y^2 + 3y - 4 = 0$$

$$(y+4)(y-1) = 0$$

$$y = -4 \quad \text{OR} \quad y = 1$$

$$x^2 = -4$$

X

$$x^2 = 1$$

$$x = \pm 1$$

$$b) 2^{2x} + 2^{x+1} - 3 = 0$$

$$(2^x)^2 + 2 \cdot (2^x) - 3 = 0$$

$$2^x = y$$

$$y^2 + 2y - 3 = 0$$

$$(y+3)(y-1) = 0$$

$$y = -3$$

$$2^x = -3$$

X

$$y = 1$$

$$2^x = 1$$

$$x = 0$$

$$(x+1)(x+2)(x+3)(x+4) = 24$$

$$(x^2 + 5x + 4)(x^2 + 5x + 6) = 24$$

$$x^2 + 5x = y$$

$$(y+4)(y+6) = 24$$

$$y^2 + 6y + 4y + \cancel{24} = \cancel{24}$$

$$y^2 + 10y = 0$$

$$y(y+10) = 0$$

$$y = 0$$

$$x^2 + 5x = 0$$

$$x(x+5) = 0$$

$$x = 0, x = -5$$

$$y = -10$$

$$x^2 + 5x = -10$$

$$x^2 + 5x + 10 = 0$$

$$D = -15$$

X

$$(a+b-2c)x^2 + (b+c-2a)x + (c+a-2b) = 0$$

$$x=1, \quad \cancel{(a+b-2c)} + \cancel{(b+c-2a)} + \cancel{(c+a-2b)} = 0$$

$$0 = 0$$

$x=1$ is a root.

$x=\beta$ is 2nd root.

$$\text{product} = \frac{c+a-2b}{a+b-2c}$$

$$\beta \cdot 1 = \frac{c+a-2b}{a+b-2c}$$

$$\beta = \frac{c+a-2b}{a+b-2c}$$

$$x=1$$

$$x = \frac{c+a-2b}{a+b-2c}$$

$$x^4 + 2x^3 - 6x^2 + 2x + 1 = 0$$

$$x^2 + 2x - 6 + \frac{2}{x} + \frac{1}{x^2} = 0$$

$$\left(x^2 + \frac{1}{x^2}\right) + \left(2x + \frac{2}{x}\right) - 6 = 0$$

$$\left(x^2 + \frac{1}{x^2}\right) + 2\left(x + \frac{1}{x}\right) - 6 = 0$$

$$x + \frac{1}{x} = y$$

$$x^2 + \frac{1}{x^2} + 2 = y^2$$

$$x^2 + \frac{1}{x^2} = y^2 - 2$$

$$(y^2 - 2) + 2y - 6 = 0$$

$$y^2 + 2y - 8 = 0$$

$$(y+4)(y-2) = 0$$

$$y = -4, \quad y = 2$$

$$y = -4$$

$$x + \frac{1}{x} = -4$$

$$\frac{x^2 + 1}{x} = -4$$

$$x^2 + 1 = -4x$$

$$x^2 + 4x + 1 = 0$$

$$x = \frac{-4 \pm \sqrt{16-4}}{2}$$

$$= \frac{-4 \pm \sqrt{12}}{2}$$

$$= \frac{-4 \pm 2\sqrt{3}}{2}$$

$$= -2 \pm \sqrt{3}$$

$$y = 2$$

$$x + \frac{1}{x} = 2$$

$$\frac{x^2 + 1}{x} = 2$$

$$x^2 + 1 = 2x$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)^2 = 0$$

$$x = 1$$

roots of the eq satisfy the eq

$$ax^2+bx+c=0 \begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$$

$$a\alpha^2+b\alpha+c=0$$

$$a\beta^2+b\beta+c=0$$

Q 96 $x=3$ is root of $x^3+2x^2-3x+a=0$, then find $a=?$

Soln

$$3^3+2\cdot 3^2-3\cdot 3+a=0$$

$$a=-36$$

Relation between roots and coeff

$$ax^2+bx+c=0 \begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$$

$$\alpha+\beta = -\frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

$$ax^2+bx+c=0 \begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$$

$$|\alpha-\beta| = \sqrt{(\alpha-\beta)^2}$$

$$= \sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$$

$$= \sqrt{\frac{b^2}{a^2} - \frac{4c}{a}}$$

$$= \sqrt{\frac{b^2-4ac}{a^2}}$$

$$= \frac{\sqrt{b^2-4ac}}{|a|}$$

$$= \frac{\sqrt{D}}{|a|}$$

$$\sqrt{x^2} = |x|$$

Q If roots of the eq $x^2 - px + (2p-2) = 0$ are two consecutive integers, find value of p ?

Soln Since two consecutive integers,

$$|\alpha - \beta| = 1$$

$$\Rightarrow \frac{\sqrt{D}}{|a|} = 1$$

$$\Rightarrow \frac{\sqrt{(-p)^2 - 4 \cdot 1 \cdot (2p-2)}}{1} = 1$$

$$\Rightarrow \sqrt{p^2 - 8p + 8} = 1$$

$$\Rightarrow p^2 - 8p + 8 = 1$$

$$\Rightarrow p^2 - 8p + 7 = 0$$

$$\Rightarrow (p-1)(p-7) = 0$$

$$p = 1, p = 7$$

B: 05 PM

Q If roots of $ax^2+bx+c=0$ are α and β , then find value of

(i) $\alpha^2 + \beta^2$

(ii) $\alpha^3 + \beta^3$

iii) $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$

Soln
 $\alpha + \beta = -\frac{b}{a}$
 $\alpha\beta = \frac{c}{a}$

$$\begin{aligned} \text{(i) } \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= \left(-\frac{b}{a}\right)^2 - 2 \cdot \frac{c}{a} \\ &= \frac{b^2}{a^2} - \frac{2c}{a} \\ &= \frac{b^2 - 2ac}{a^2} \end{aligned}$$

$$\begin{aligned} \text{(ii) } \alpha^3 + \beta^3 &= (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) \\ &= -\frac{b}{a} \left[(\alpha + \beta)^2 - 3\alpha\beta \right] \\ &= -\frac{b}{a} \left[\left(-\frac{b}{a}\right)^2 - 3 \frac{c}{a} \right] \\ &= -\frac{b}{a} \times \left[\frac{b^2 - 3ac}{a^2} \right] \\ &= \frac{-b(b^2 - 3ac)}{a^3} \end{aligned}$$

$$\text{iii)} \quad \frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$$

$$\Rightarrow \frac{1}{\left(-\frac{c}{a}\right)^2} + \frac{1}{\left(-\frac{c}{a}\right)^2}$$

$$= \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2}$$

$$= \frac{\alpha^2 + \beta^2}{c^2}$$

$$ax^2 + bx + c = 0 \quad \alpha, \beta$$

$$a\alpha^2 + b\alpha + c = 0$$

$$\alpha(a\alpha + b) + c = 0$$

$$a\alpha + b = -\frac{c}{\alpha}$$

Q 9 If α and β are roots of $x^2 - 3x - 7 = 0$, then find value of $\frac{\alpha^5 - 7\alpha^3}{3\alpha + 7} + \frac{\beta^5 - 7\beta^3}{3\beta + 7}$

$$\begin{aligned}
& \frac{\alpha^5 - 7\alpha^3}{3\alpha + 7} + \frac{\beta^5 - 7\beta^3}{3\beta + 7} \\
&= \frac{\alpha^3(\alpha^2 - 7)}{3\alpha + 7} + \frac{\beta^3(\beta^2 - 7)}{3\beta + 7} \\
&= \frac{\alpha^3(3\alpha)}{\alpha^2} + \frac{\beta^3(3\beta)}{\beta^2} \\
&= 3\alpha^2 + 3\beta^2 \\
&= 3[(\alpha + \beta)^2 - 2\alpha\beta] \\
&= 3[9 + 14] \\
&= 69
\end{aligned}$$

$$\begin{cases}
\alpha^2 - 3\alpha - 7 = 0 \\
3\alpha + 7 = \alpha^2 \\
\alpha^2 - 7 = 3\alpha
\end{cases}$$

$$\alpha + \beta = 3, \quad \alpha\beta = -7$$

Newton's method

Q If roots of the eq $x^2 - 6x - 2 = 0$ are α and β and $a_n = \alpha^n + \beta^n$, then find values of $\frac{a_{10} - 2a_8}{3a_9} = ?$

Soln

$$\frac{a_{10} - 2a_8}{3a_9} = \frac{(\alpha^{10} + \beta^{10}) - 2(\alpha^8 + \beta^8)}{3(\alpha^9 + \beta^9)}$$
$$= \frac{\alpha^{10} - 2\alpha^8 + \beta^{10} - 2\beta^8}{3(\alpha^9 + \beta^9)}$$
$$= \frac{\alpha^8(\alpha^2 - 2) + \beta^8(\beta^2 - 2)}{3(\alpha^9 + \beta^9)}$$
$$= \frac{6\alpha^9 + 6\beta^9}{3(\alpha^9 + \beta^9)} = 2$$

$\alpha^2 - 6\alpha - 2 = 0$ $\alpha^2 - 2 = 6\alpha$

}

Q If roots of the eq $x^2 - 7x - 1 = 0$ are α and β and $a_n = \alpha^n + \beta^n$ then find value of $\frac{a_{14} + a_{10}}{a_{12}} = ?$

$$\begin{aligned}
& \frac{a_{14} + a_{10}}{a_{12}} \\
&= \frac{\alpha^{14} + \beta^{14} + \alpha^{10} + \beta^{10}}{\alpha^{12} + \beta^{12}} \\
&= \frac{\alpha^{14} + \alpha^{10} + \beta^{14} + \beta^{10}}{\alpha^{12} + \beta^{12}} \\
&= \frac{\alpha^{10}(\alpha^4 + 1) + \beta^{10}(\beta^4 + 1)}{\alpha^{12} + \beta^{12}} \\
&= \frac{51\alpha^{12} + 51\beta^{12}}{\alpha^{12} + \beta^{12}} \\
&= 51
\end{aligned}$$

$$\alpha^2 - 7\alpha - 1 = 0$$

$$\alpha^2 - 1 = 7\alpha$$

$$(\alpha^2 - 1)^2 = 49\alpha^2$$

$$\alpha^4 + 1 - 2\alpha^2 = 49\alpha^2$$

$$\alpha^4 + 1 = 51\alpha^2$$

Q If one root of the eq $x^2 + px + 8 = 0$ is square of the other
find value of p .

Q If roots of the eq $x^2 - ax + b = 0$ are in the ratio $m:n$,
find relationship between a, b, m, n .

Soln roots are α and α^2

$$\begin{array}{l|l} \alpha + \alpha^2 = -p & \alpha \cdot \alpha^2 = 8 \\ 2 + 4 = -p & \alpha^3 = 8 \\ p = -6 & \alpha = 2 \end{array}$$

Soln

$$x^2 - ax + b = 0$$

roots are $m\alpha$ and $n\alpha$

$$m\alpha + n\alpha = \frac{-(-a)}{1}$$

$$(m+n)\alpha = a$$

$$\therefore \alpha = \frac{a}{m+n}$$

$$m\alpha \cdot n\alpha = \frac{b}{1}$$

$$mn\alpha^2 = b$$

$$mn \left(\frac{a}{m+n} \right)^2 = b$$

$$mn a^2 = b(m+n)^2$$

Q 96 $ap^2 + bp + cp + ab + ac = 0$ and $aq^2 + bq + cq + ab + ac = 0$

and $p \neq q$, then find $pq = ?$

$$ap^2 + bp + cp + ab + ac = 0$$

$$aq^2 + bq + cq + ab + ac = 0$$

$$ax^2 + bx + cx + (ab + ac) = 0 \begin{matrix} \nearrow p \\ \searrow q \end{matrix}$$

$$pq = \frac{ab + ac}{a}$$

$$= b + c$$